

Practical undecidability problems.

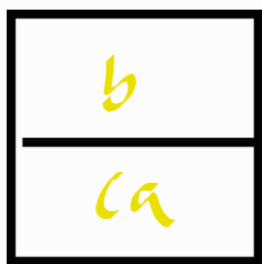
Post correspondence problem (PCP)

- you are given a collection of domino cards, with string written below and above the mid-line at each card.

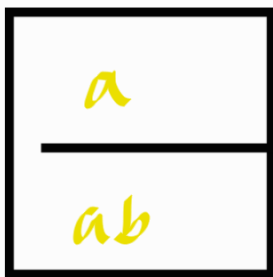
abc	aa	bac
ab	ba	abc

- List these cards one by one, so that the strings above match strings below.
- You can use the same card any number of times including zero times.
- You can think of this as having infinite copies of each domino.

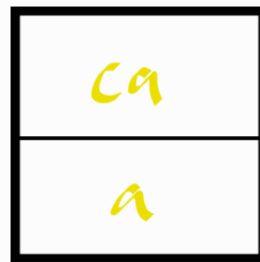
Ex: {



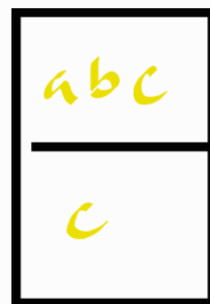
card 1



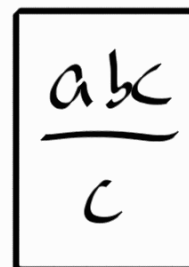
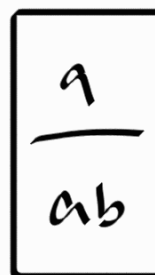
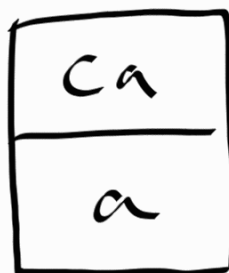
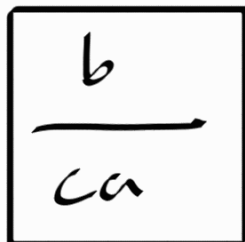
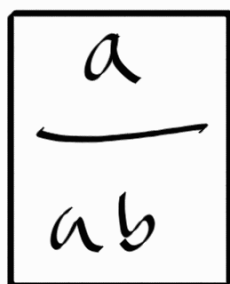
card 2



card 3

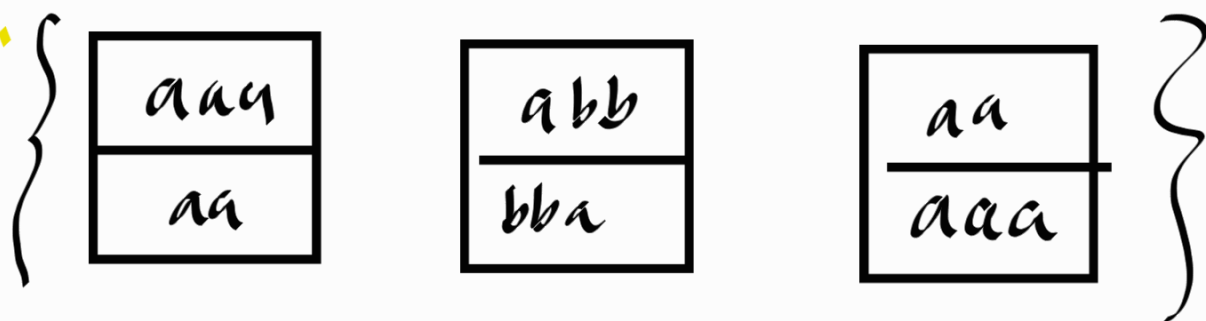


card 4 }



$$\begin{array}{r}
 abcaabc \\
 \hline
 abcaabc
 \end{array}$$

Ex 2'



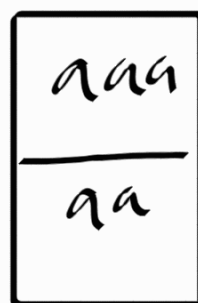
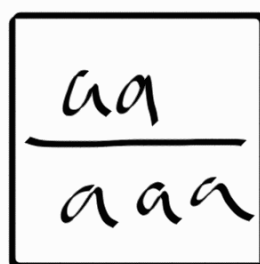
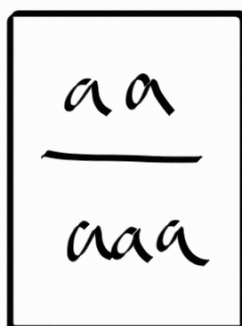
1

2

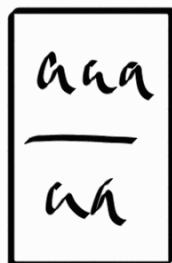
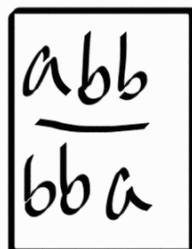
3

$\langle 1, 3 \rangle$

$\langle 3, 1 \rangle$



$\langle 3, 2, 1 \rangle$



Ex 3:

bb
b

1

bbq
aa

2

abbb
baaa

3

No solutions

$$P = \{ \langle D_1 \rangle, \langle D_2 \rangle, \dots, \langle D_k \rangle \}$$

$$PCP = \{ \langle P \rangle \mid P \text{ is an instance of Post correspondence problem with a match} \}$$

The PCP is undecidable

Proof is too long

Please read the proof in the book,

Mapping Reducibility

— Map Reductions $\leq, \alpha, \leq_m$

Definitions

A function $f: \Sigma^* \rightarrow \Sigma^*$ is a computable function if some TM M , on every input w , halts with $f(w)$ on its tape.

A TM computes a function by starting with the input to the function on the tape and halting with the output of the function on the tape.

Ex: Arithmetic operations on integers are computable functions.

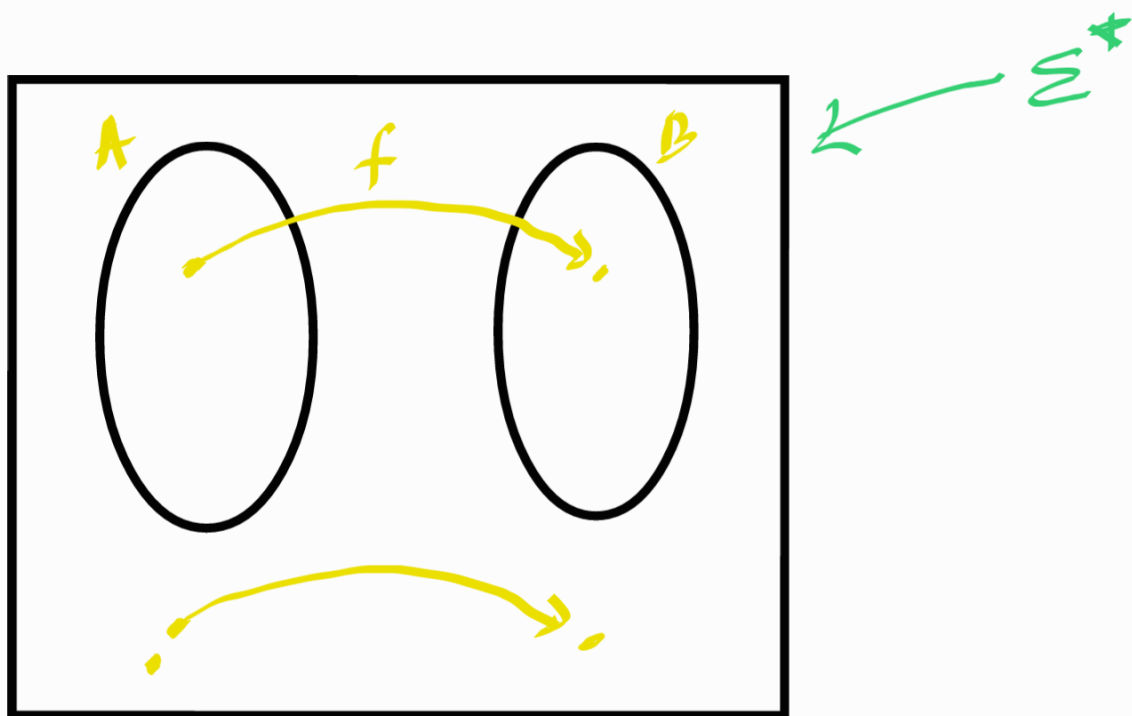
Definition

Mapping Reducibility

A language A is mapping reducible to language B , written as $A \leq_m B$, If there is a computable function $f: \Sigma^* \rightarrow \Sigma^*$, where for every w

$$w \in A \iff f(w) \in B$$

The function f is called the reduction from A to B .



All previous reductions are map reductions.

* In a map reduction, you solve an instance of A by calling a solution to B only once.

Theorem 5.22

If $A \leq_m B$ and B is decidable, then A is decidable.

Corollary 5.23

If $A \leq_m B$, and A is undecidable, then B is undecidable.

Theorem 5.28

If $A \leq_m B$ and B is Turing-recognizable, then A is Turing-recognizable.

